*In this test, you have to prepare **specification with 3-3 test cases**, small **algorithm**, and the **code** using the specification tool, the structogram tool, and the TMS system for a small problem.*

Prepare the solutions based on the patterns.

*In the postcondition (of the specification), you should use the short form, and the algorithm needs to follow your specification.The **links to the specification and algorithm** should be at the **beginning of your code as a comment**.Your test can be downloaded from the **TMS system**, and you should test and upload your code (with the links to the specification and the algorithm) into the TMS system.*

Beavers

The Mura-Drava-Danube Biosphere Reserve conducted a study on the beaver population across "p" locations.

Write a program that checks whether each location has at least 10 beavers.

Input

The first line of the *standard input* contains the number of the places (1≤p≤100). The next p lines contain the number of the beavers (0≤beavers[i]≤20) in each place.

Output

The first line of the *standard output* should contain a single text "true" or "false" based on the answer.

Example₁

Input	*Output*
3 2 15 10	false

Example₂

Input	*Output*
3 11 12 22	yes

Solution

Specification

In: p∈N, beavers∈N[1..p]

Out: eloc∈L

Pre: ∀i∈[1..p]:(0≤beavers[i]≤20)

Post: eloc=**FORALL**(i=1..p,beavers[i]≥10)

Pattern

decision all - all=FORALL(i=b..e,A(i))

Reduction table

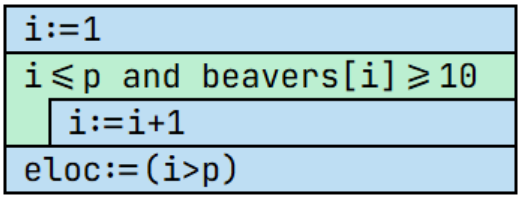
all	~	eloc
b..e	~	1..p
A(i)	~	beavers[i]≥10

Test cases

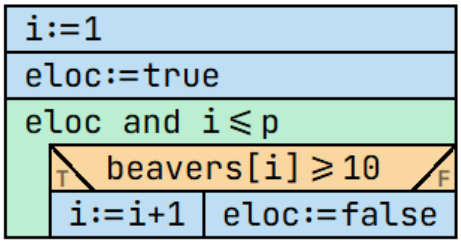
p: 0 beavers: [] eloc: true	p: 5 beavers: [11,12,12,12,12] eloc: true	p: 5 beavers: [11,12,1,12,12] eloc: false

Link of the specification, test cases and reduction table.

Algorithm



or



Link of the algorithm

Code

```
internal class Program {
   static void Main(string[] args) {
   //define variables
      int p; int[] beavers; // input
      bool eloc;            // output
   // read the input
      Console.Error.WriteLine("Number of locations (p): ");
      p = int.Parse(Console.ReadLine());
      beavers = new int[p];
      for (int i=0;i<p;i++) {
         Console.Error.WriteLine($"Number of the beavers in the {i+1}. location: ");
         beavers[i] = int.Parse(Console.ReadLine());
      }
    // solve the task
      int j = 0;
      while (j<p && beavers[j]>=10) { j++; }
      eloc = (j >= p);
   // write the output
      Console.WriteLine(eloc);
   }
}
```